Ground Reference Information

Reference Data/Information

- **The acquisition of reference data involves collecting measurements or observations about the objects, areas, or phenomena that are being or have been remotely sensed,**

  "Ground Truth"vs. "Ground Reference"

- **Design Objectives for Ground Information Collection**
  - **What is the end goal?**
  - **What information is necessary?**

Purposes of Reference Data

- **To aid in the analysis and interpretation of remotely sensed data (e.g. classification results)**
- **To verify information extracted from remote sensing data (e.g. accuracy assessment)**

Data Integrity

- **Data Integrity**
- **Data integration**
  - **Scale, resolution**
- **Data interpretation and analysis**
  - **Inferences, accurate classification**
- **Quantify data for application**
  - **Area, location**
- **Verify information or calibrate**
  - **Sensor or measurement errors**

Purposes of Reference Data

- **For designing and calibrating a sensor**
- **To aid in the analysis and interpretation of remotely sensed data (e.g., classification results)**
- **To verify information extracted from remote sensing data (e.g., accuracy assessment)**

Types of Reference Data

- **Maps : Relief Maps, Soil Surveys Maps, Contour Maps**
  - **A Good Starting Point**
  - **Scaled to Match Many Resolutions**
  - **Readily Available**
- **Descriptive reports and inventory tallies**
- **Aerial photography, satellite imagery, DEM/DSM,DOQ**
- **In situ measurements**
- **Ground reference information such as type of crops, stage of growth, Locals, Field history**

Agricultural Ground Reference Data

- Crop type
- Crop variety
- Plant population
- Residue
- Growth stage
- Row spacing/direction
- Yield
- Previous crop

- Planting Techniques
- Date of Planting
- Date of Harvest
- Type of Harvesting Method
- Species Variety
- Agricultural Patterns
- Soils
- Moisture Content
- Temperature
- Precipitation

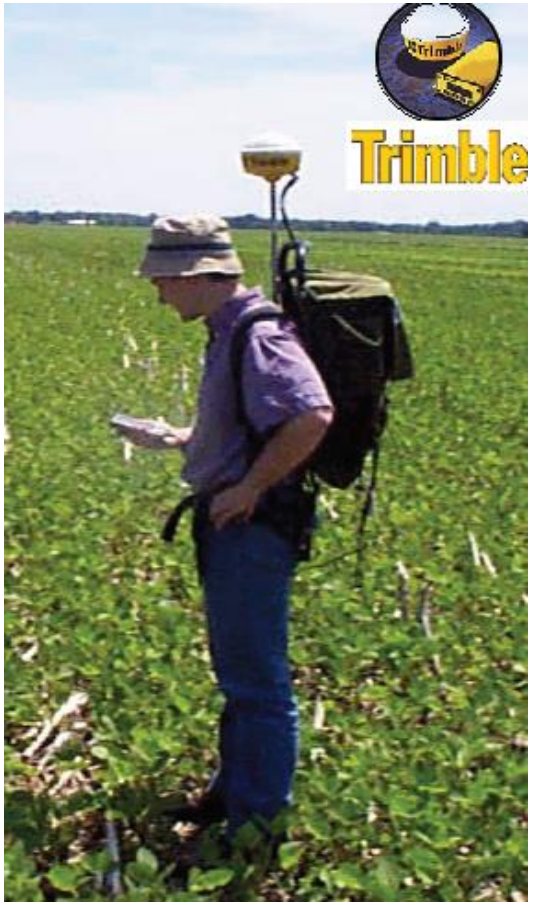
Assessment of Site Suitability

- Topographic Relief
- Slope Stability
- Bearing Strength of Soils
- Extent of Forest Cover
- Land Ownership
- Cost of Available Land
- Available Water
- Government Regulations
- Direction of Prevailing Winds
- Commodity Prices
- Soil Productivity
- Slope
- Flooding Potential

Selection of land use choices will base on a variety of reasons

Geo referenced Data

- Recording observations/measurements –GPS
- Yield monitor data
- Remotely sensed imagery
  - Locate areas of interest within a field
  - Aid in quantifying potential areas of control
  - Aid in analysis





Taking Ground Photographs



View from the camera



Camera on a pole

Summary

- **Data should reflect or be homogeneous with level of precision of the end goal**
- **Accurate representation of the “within field spatial variability” should exist**
- **Accuracy of information is only as good as the technology and data used for research/application**
- **Most of time, people collect the wrong reference data, information**
- **Planning helps to collect the “right data and information”**